

CHE110:ENVIRONMENTAL STUDIES

L:2 T:0 P:0 Credits:2

Course Outcomes: Through this course students should be able to

CO1 :: define current environmental issues and associated problems.

CO2 :: explain various components of environment and their role in maintaining the ecosystem and biodiversity.

CO3 :: determine the role of various environmental policies, practices and techniques to reduce environmental pollution.

CO4 :: analyze various environmental phenomena and their relationship with human communities.

Unit I

Introduction and sustainable development : introduction to environment, components of environment, spheres of earth, multidisciplinary nature, scope and importance of environmental studies, concept of sustainability, sustainable development, carrying capacity, pillars of sustainability, sustainable development goals

Unit II

Natural resources and ecosystem : introduction and classification, renewable and non renewable resources, land resources, land degradation, soil erosion and desertification, forest resources, causes and impacts of deforestation, water resources, use and over-exploitation of surface and ground water, conflicts over water, energy resources, use of alternate energy sources, growing energy needs, conservation of natural resources and human role, ecosystem: structure and function of ecosystem, types of ecosystem, energy flow in an ecosystem: food chains and food webs, ecological pyramids, ecological succession, Case studies of the following ecosystems : a)forest ecosystem b) grassland ecosystem c) desert ecosystem d) aquatic ecosystem

Unit III

Biodiversity and conservation : definition, levels of biological diversity: genetic, species and ecosystem diversity, importance of biodiversity:ecosystem and biodiversity services, ecological, economic, social, aesthetic and informational value, hot spots of biodiversity, threats to biodiversity: habitat loss, poaching of wildlife, biological invasions, human wildlife conflicts, endangered and endemic species in India, conservation of biodiversity: in-situ and ex-situ conservation methods of biodiversity, biogeographic zones of India, India as a mega diversity nation

Unit IV

Environmental pollution : definition, types and causes, effects and controls: air, water, soil, noise and radiation pollution, emerging pollutants, fireworks and their ill-effects, climate change, global warming, ozone layer depletion, acid rain and impacts on human communities and agriculture, case studies on pollution

Unit V

Environmental policies and practices : environmental laws: Wildlife Protection Act, Water (Prevention and control of Pollution) Act, Forest Conservation Act, Air (Prevention & Control of Pollution) Act, Environmental Protection Act, Environmental Impact Assessment, International agreements: Montreal and Kyoto protocols, Convention on Biological Diversity (CBD), solid waste management: control measures of urban and industrial waste, role of tribal populations in policy making

Unit VI

Human communities and environment : human population growth: impacts on environment, human health and welfare, environmental movements: Chipko movement, Silent valley movement, Appiko movement, Jungle bachao andolan, Narmada Bachao Andolan, Tehri Dam conflict, Bishnois of Rajasthan, role of Indian and other religions and cultures in environmental conservation, environmental ethics, public awareness, disaster management, floods, droughts, earthquake, landslides and cyclones

Text Books:

1. PERSPECTIVE IN ENVIRONMENTAL STUDIES by ANUBHA KAUSHIK, C P KAUSHIK, NEW AGE INTERNATIONAL PUBLISHERS

References:

1. TEXT BOOK OF ENVIRONMENTAL STUDIES by D. DAVE AND S. S. KATEWA, CENGAGE LEARNING

